ADVICE DOCUMENT

PlaySafe

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1.Introduction

This document gives advice on how to build PlaySafe, a model that estimates the probability that a LaLiga player will have at least one injury in the next season. It is written for the technical staff of a football club and for the project team who will build the model.

It explains the context and its uncertainty, recommends the data, methods and tools to use, and describes the ethical and sustainability criteria to consider. It is written before the final model is built, so it is a recommendation and not a result.

2.Purpose and audience

This document advises the stakeholders of PlaySafe on which data-driven solution to build, with which data, methods and tools, and under which ethical and sustainability conditions.

- **Main stakeholders:** sports scientists, physical trainers and medical staff of LaLiga clubs. They can use the estimates to adapt training.
- **Other stakeholders:** technical directors and analysts, who plan the squad before a season or a transfer window.

3.Recommendations

3.1 Recommended solution

We recommend a model that estimates the probability that a player has at least one injury in the next season. The result is a probability, for example 72%, and not only a yes or no, because the staff can use it to rank players and decide where to look first.

This is a supervised binary classification problem: the model learns from past seasons in which we know if the player was injured in the following one. We recommend a simple solution because the data is limited and the staff must be able to understand and trust the result. We are not able to predict the date or the severity of injuries, because the data does not allow it.

3.2 Recommended data

- **Source:** the laliga_players.csv dataset from the football-injury-scraper repository (3,713 injury records of 490 players, seasons 12/13 to 24/25). It is the best source available for this project, but it is not official, so its completeness cannot be guaranteed.
- **Variables:** player characteristics (age in each season, not the current age, height and position), playing time (minutes played) and injury history (number of injuries, days out and matches missed).

- **Structure:** one row per player and season, with the target "injury in the next season" .
- **Quality checks:** convert the formats, treat special values such as "-" and "?" as missing, fill the missing values and check the names of the players.
- **No data leakage:** use only information from before the season that is predicted.
- **Missing data:** training load (GPS), medical information, and the type and dates of the injuries.

3.3 Recommended methods and tools

Choice	Recommendation	Reason
Models	Logistic Regression as a baseline and Random Forest to compare	Both give a probability. Logistic Regression is simple to explain, and Random Forest can find more complex patterns and shows which variables matter.
Models not recommended	XGBoost and neural networks	They are too complex and harder to explain for this project.
Validation	Time-based split: train with earlier seasons and test with later seasons	In real life the future is predicted from the past, and this avoids data leakage.
Metrics	Accuracy, precision, recall and F1-score, with recall as the priority	The aim is to find as many players at risk as possible. The results must be compared with a model that only guesses.
Tools	Python with pandas and scikit-learn, in Jupyter notebooks	They are free, well documented, light and easy to use.
Prototype	A simple notebook demo. As future improvement it can be converted to an application	The club will not work with notebooks.

4. Risks and conditions

The main risks are:

- The model can be wrong in many cases, because injuries have many causes.
- The data can be incomplete or have biases.
- The same players appear in past and future seasons, so the model may recognise players instead of learning general patterns.
- The estimates can be used in a wrong way, for example for contracts or selection.